T.Y.B.Sc. (I.T.) — SEMESTER V

DATA VISUALIZATION WITH TABLEAU

Q Bank Practicals — Steps, Expected Outputs and Diagrams

Based on the uploaded tableau-1.docx. The original sub-question structure and steps are retained.

Dataset used for most chart practicals: Sample – Superstore. Diagrams below are illustrative expected outputs, not screenshots from Tableau; actual values and appearance may vary with your dataset and Tableau version.

Q1. a) Create stacked and grouped bar charts for comparison.

Steps

1. Grouped Bar Chart
2. Open Sample – Superstore.
3. Drag Category → Columns.
4. Drag Sales → Rows.
5. Drag Region → Columns, beside Category.
6. Stacked Bar Chart
7. Create a new worksheet.
8. Drag Category → Columns.
9. Drag Sales → Rows.
10. Drag Region → Marks → Color.
11. Set Marks type to Bar.

Expected Output: Two comparison views are available: grouped bars show Region bars side by side for each Category; stacked bars show each Region as a colored segment within the Category bar.

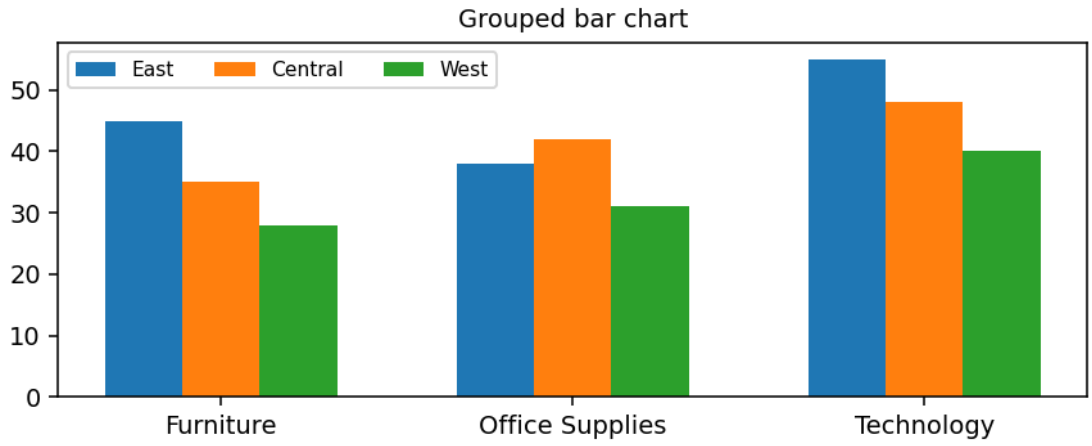
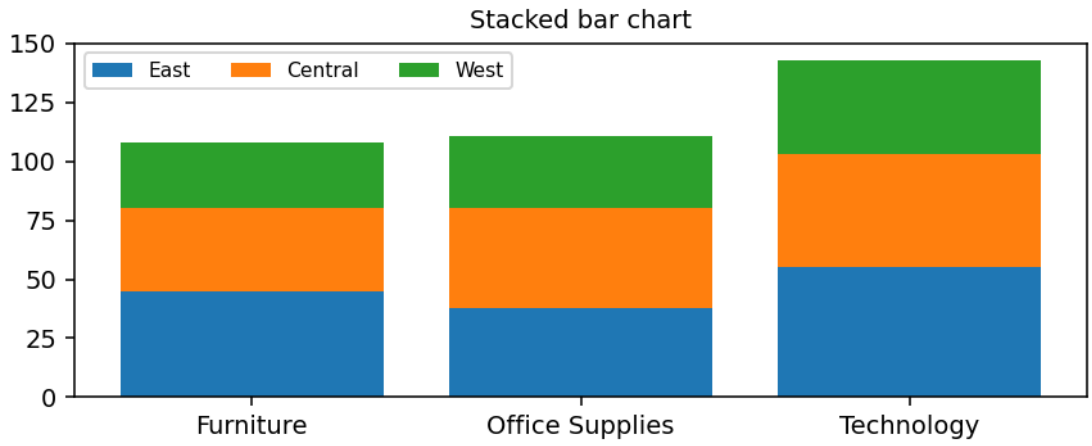


Diagram: illustrative expected output



Expected output for stacked bar chart

Q1. b) Build treemaps and pie charts for part-to-whole analysis.

Steps

12. Treemap
13. Drag Category → Label.
14. Drag Sales → Size.
15. Drag Category → Color.
16. Select Treemap from Show Me.
17. Pie Chart
18. Set Marks type to Pie.
19. Drag Category → Color.
20. Drag Sales → Angle.
21. Drag Category and Sales → Label.

Expected Output: A treemap displays rectangles by category, sized by Sales. The pie chart shows category slices and sales labels, making each category's share visible.

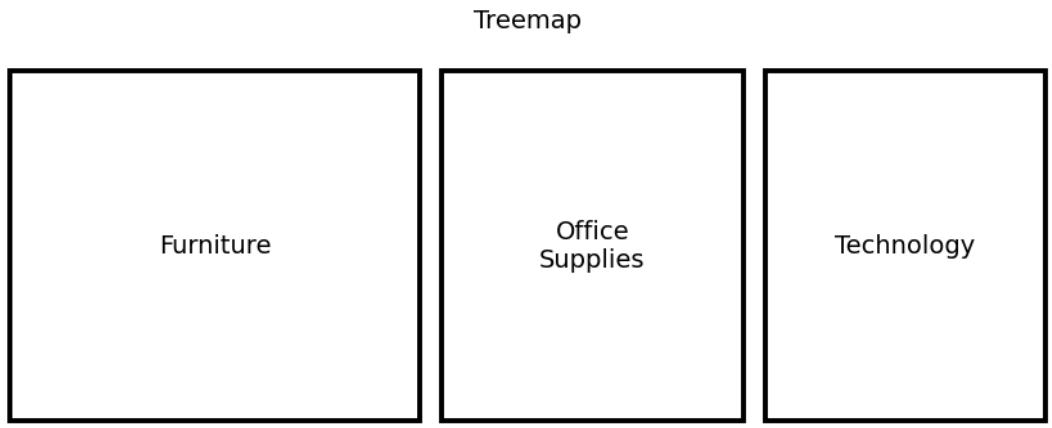
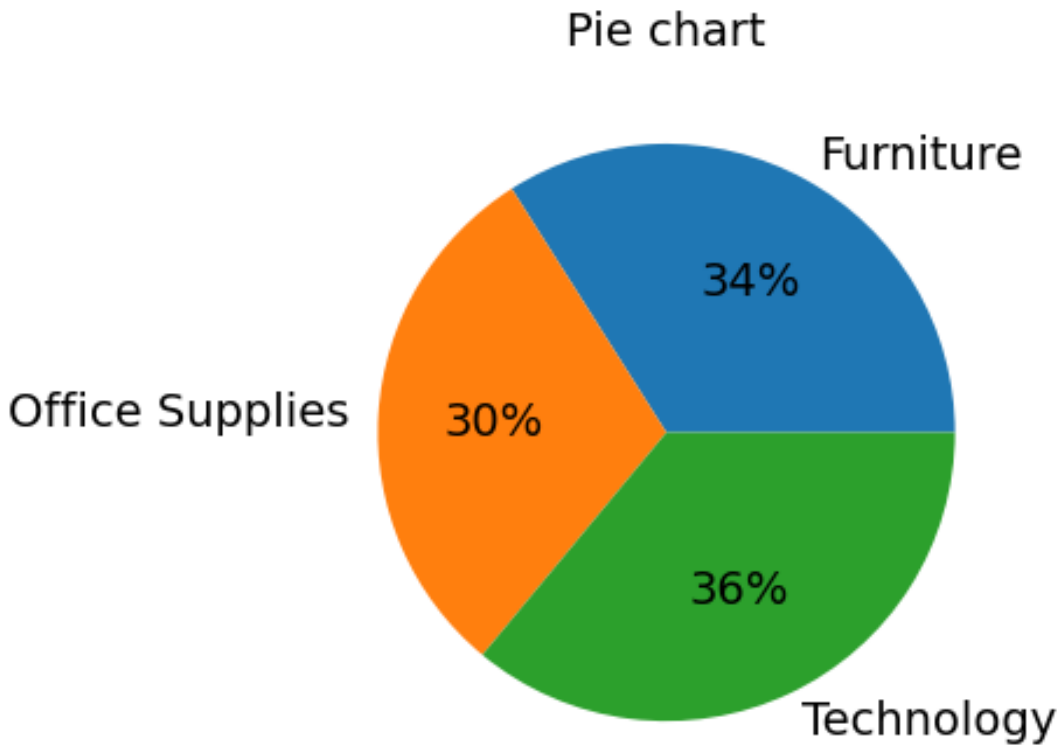


Diagram: illustrative expected output



Expected output for pie chart

Q1. c) Manage metadata (rename, hide, change data types).

Steps

22. Rename
23. Right-click Sales in the Data pane.
24. Select Rename.
25. Type Total Sales → Enter.
26. Hide
27. Right-click the required field.
28. Select Hide.
29. Change Data Type
30. Right-click the field.
31. Select Change Data Type.
32. Choose String, Number, Date, Date & Time, or Boolean.

Expected Output: The Data pane shows the renamed field as Total Sales; the hidden field no longer appears in the normal field list; the chosen field icon/type changes to the selected data type.

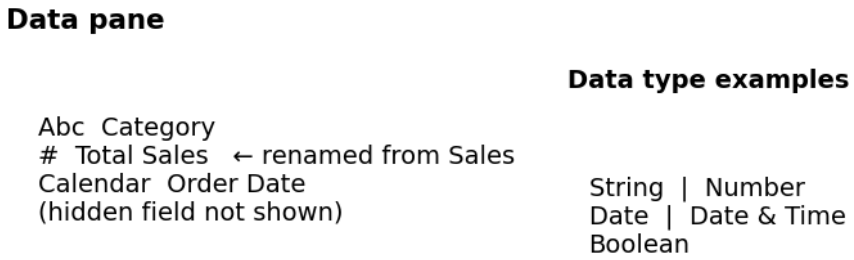


Diagram: illustrative expected output

Q2. a) Design a basic dashboard layout using sheets.

Steps

33. Create Sheet 1: Category → Columns, Sales → Rows.
34. Create Sheet 2: Region → Columns, Sales → Rows.
35. Click New Dashboard.
36. Drag both sheets onto the dashboard.
37. Enable Show Dashboard Title.
38. Rename it Sales Dashboard.

Expected Output: A dashboard titled Sales Dashboard contains two views: Sales by Category and Sales by Region.

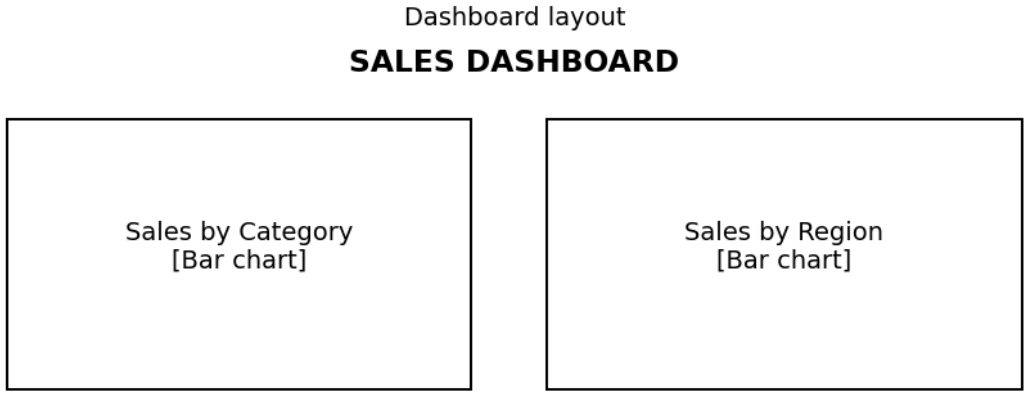


Diagram: illustrative expected output

Q2. b) Add filters and interactive controls.

Steps

39. Drag Region → Filters.
40. Select All → OK.
41. Right-click the filter → Show Filter.
42. On the dashboard, select Apply to Worksheets → All Using This Data Source.

Expected Output: A Region filter card appears on the dashboard. Choosing a region updates the sheets using the same data source.

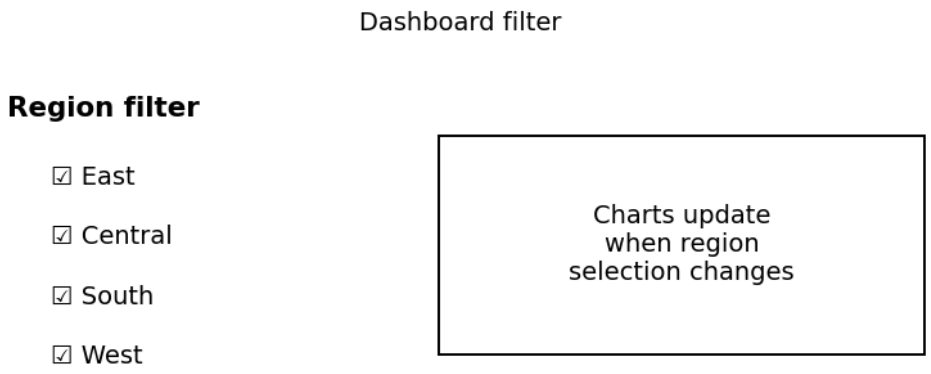


Diagram: illustrative expected output

Q3. a) Design scatter plots to identify relationships.

Steps

43. Create a new worksheet.
44. Drag Sales → Columns.
45. Drag Profit → Rows.
46. Drag Category → Color.
47. Drag Category → Detail.

Expected Output: Scatter plot showing the relationship between Sales and Profit; marks are colored by Category.

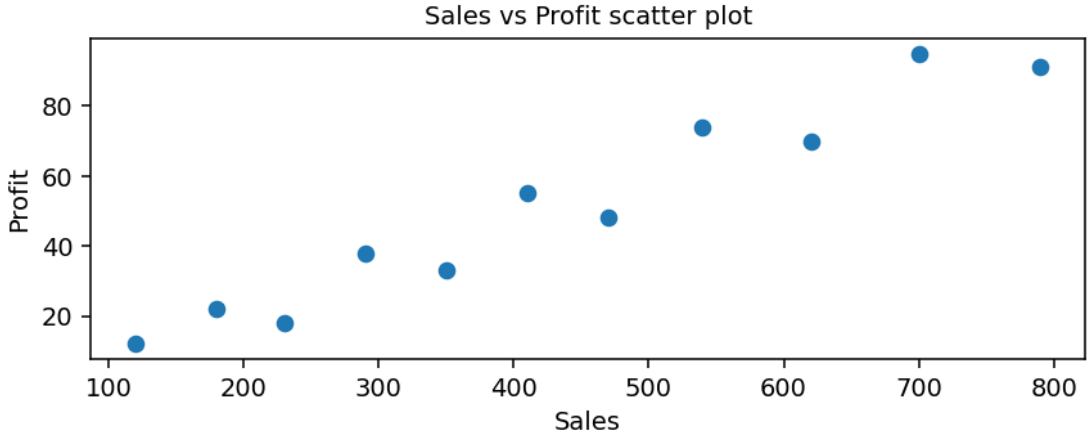


Diagram: illustrative expected output

Q3. b) Create dual-axis and combination charts.

Steps

48. Drag Order Date → Columns.
49. Drag Sales → Rows.
50. Drag Profit → Rows.
51. Right-click the second axis → Dual Axis.

52. Set Sales → Bar.
53. Set Profit → Line.

Expected Output: A combination chart shows Sales as bars and Profit as a line with two axes.

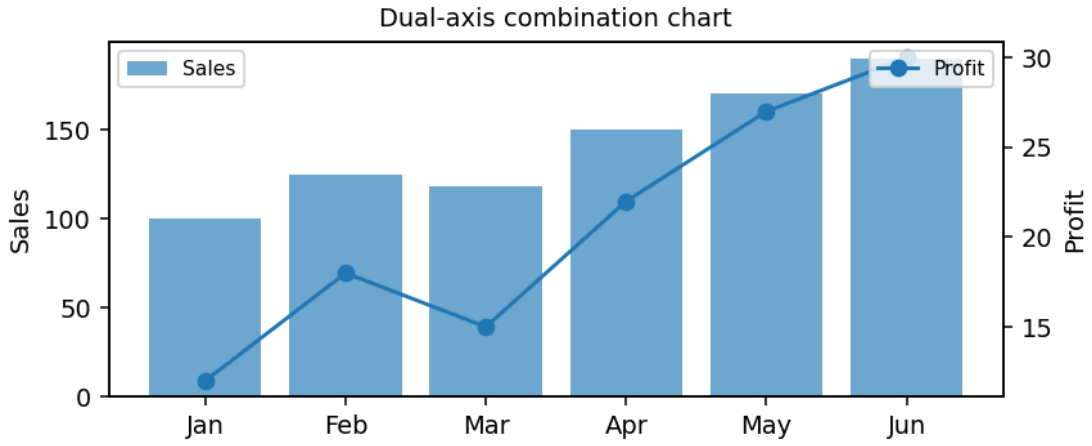


Diagram: illustrative expected output

Q3. c) Implement different types of filters (dimension, measure, date).

Steps

54. Dimension Filter
55. Drag Category → Filters.
56. Select categories → OK.
57. Measure Filter
58. Drag Sales → Filters.
59. Select aggregation and range → OK.
60. Date Filter
61. Drag Order Date → Filters.
62. Select Years, Months or Exact Date → OK.

Expected Output: Three filter types can be applied: a dimension filter selects categories, a measure filter limits values by range, and a date filter limits the time period.

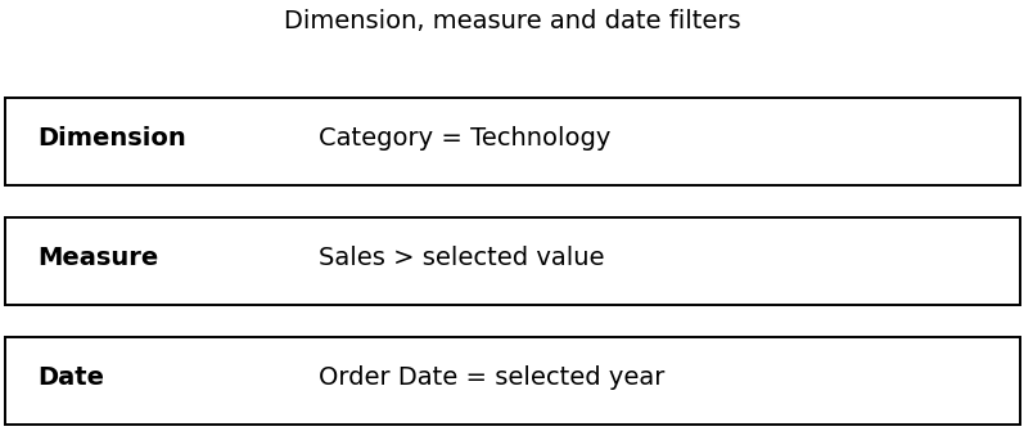


Diagram: illustrative expected output

Q4. a) Create informative tooltips.

Steps

63. Open a worksheet containing a chart.
64. Click Marks → Tooltip.
65. Enter the text shown below.
66. Click OK.

Formula / text: Sales Information\nCategory: <Category>\nSales: <SUM(Sales)>

Expected Output: When a user hovers over a mark, a tooltip appears with the category and sales information.

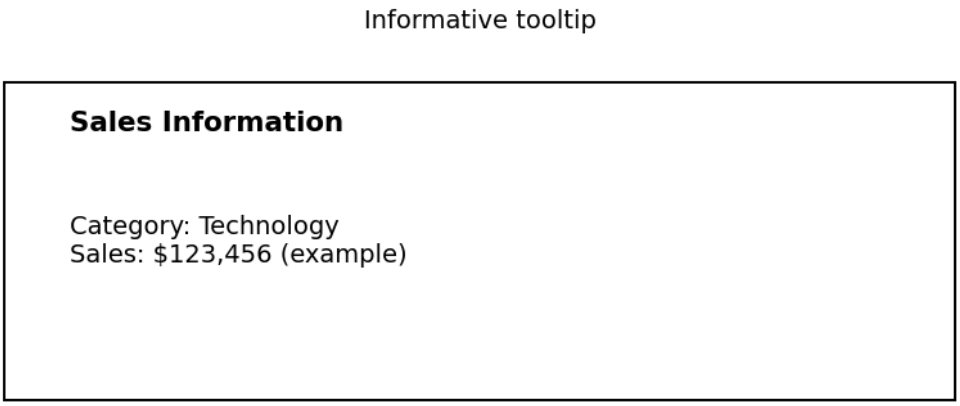


Diagram: illustrative expected output

Note: Use Tableau's Insert menu to insert field values if you prefer not to type the field placeholders manually.

Q4. b) Implement “Viz in Tooltip”.

Steps

67. Create main chart: Sales by Category.
68. Create another worksheet: Sales Details.
69. Create a chart using Sub-Category and Sales.
70. Return to the main worksheet.
71. Click Marks → Tooltip → Insert → Sheets.
72. Select Sales Details → OK.

Expected Output: A small Sales Details chart appears inside the tooltip when hovering over a category in the main chart.

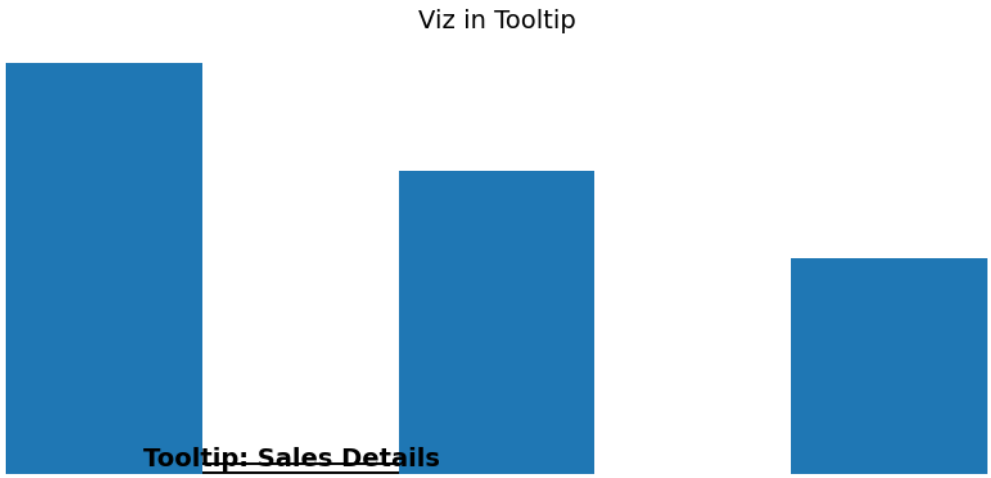


Diagram: illustrative expected output

Q5. a) Build box-and-whisker plots.

Steps

73. Create a new worksheet.
74. Drag Category → Columns.
75. Drag Sales → Rows.
76. Open Show Me.
77. Select Box-and-Whisker Plot.

Expected Output: A box-and-whisker plot appears for each Category, showing the median, quartiles, spread and possible outliers.



Diagram: illustrative expected output

Q5. b) Develop histograms for distribution analysis.

Steps

78. Create a new worksheet.
79. Select Sales in the Data pane.
80. Open Show Me.
81. Select Histogram.

Expected Output: A histogram groups Sales values into bins and shows the frequency of records in each bin.

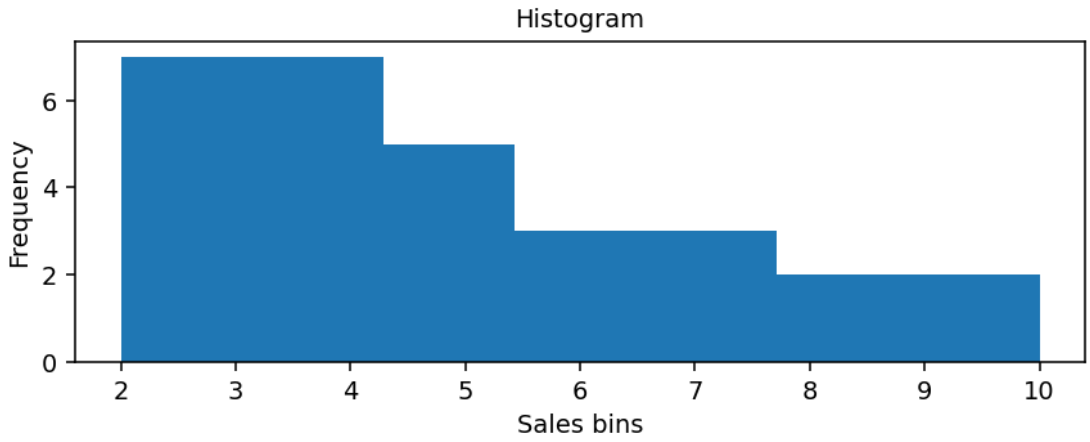


Diagram: illustrative expected output

Q5. c) Perform joins (inner, left, right) on multiple tables.

Steps

82. Open the Data Source page.
83. Add the first table.
84. Drag the second table beside it.
85. Select the common field, such as Customer ID.
86. Choose the required join type.
87. Inner Join: Matching records only.
88. Left Join: All left records + matching right records.
89. Right Join: All right records + matching left records.

Expected Output: The data preview displays the combined table. The records retained depend on whether Inner, Left or Right Join is selected.

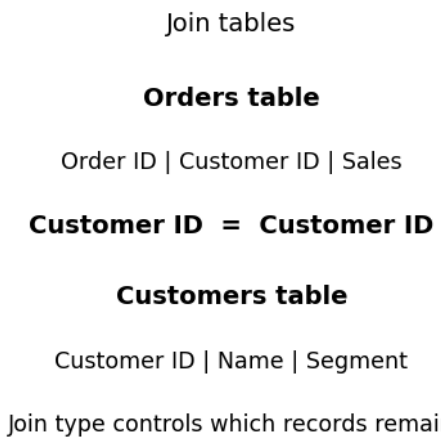


Diagram: illustrative expected output

Q6. a) Apply worksheet-level and workbook-level formatting.

Steps

90. Worksheet-Level Formatting

91. Open a worksheet.
92. Click Format.
93. Customize fonts, alignment, borders, shading and gridlines.
94. Workbook-Level Formatting
95. Click Format → Workbook, if available.
96. Customize the general formatting settings.

Expected Output: The worksheet shows the selected font, alignment, border, shading and gridline settings. Workbook-level formatting applies general settings where supported by the Tableau version.

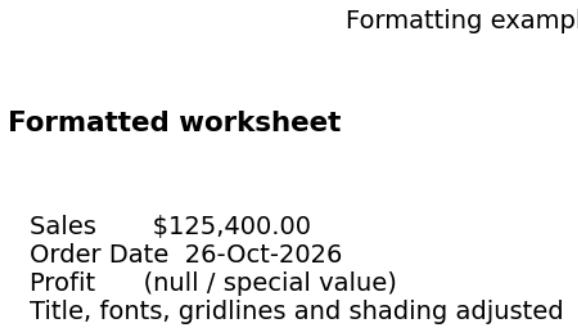


Diagram: illustrative expected output

Q6. b) Customize number, date, and null value formatting.

Steps

97. Number
98. Right-click Sales.
99. Select Format.
100. Choose number/currency format.
101. Date
102. Right-click Order Date.
103. Select Format.
104. Choose the required date format.
105. Null
106. Locate null/special values.
107. Use the available special-value formatting options.

Expected Output: Sales appears in the selected number or currency format, Order Date uses the selected date style, and null/special values are displayed or formatted using the available options.

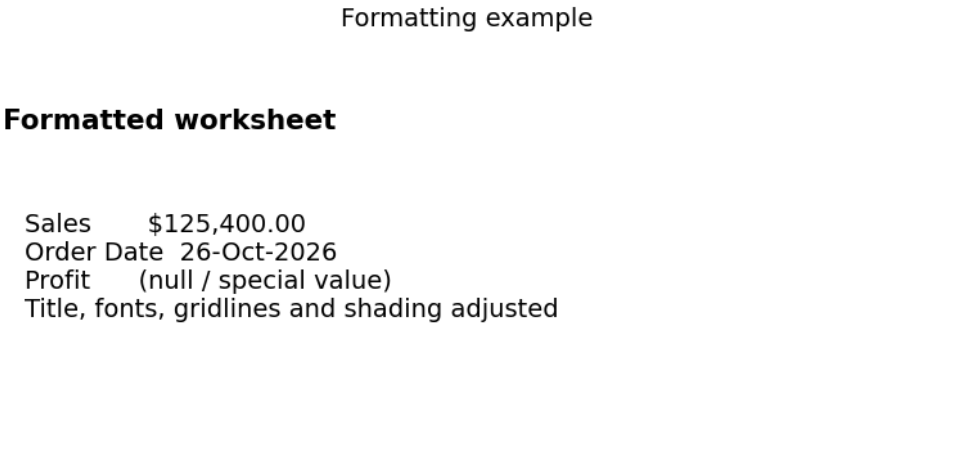


Diagram: illustrative expected output

Q7. a) Implement highlight and filter actions.

Steps

108. Highlight Action
109. Create a dashboard with two charts.
110. Click Dashboard → Actions.
111. Select Add Action → Highlight.
112. Choose source and target sheets.
113. Select the required field → OK.
114. Filter Action
115. Click Dashboard → Actions.
116. Select Add Action → Filter.
117. Choose source and target sheets.
118. Select the required field → OK.

Expected Output: Selecting a mark in the source chart highlights related marks or filters the target chart, depending on the action configured.

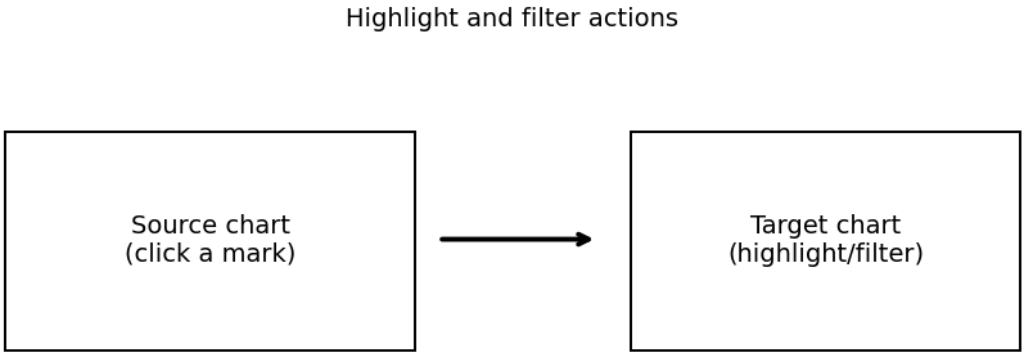


Diagram: illustrative expected output

Q7. b) Create dynamic dashboards using parameters.

Steps

119. Right-click Data pane → Create Parameter.
120. Name it Select Measure.
121. Set data type to String.
122. Add values: Sales, Profit.
123. Create calculated field: Selected Measure.
124. Enter the formula shown below.
125. Use Selected Measure in the chart.
126. Right-click the parameter → Show Parameter.

Formula / text: IF [Select Measure] = "Sales"\nTHEN [Sales]\nELSE [Profit]\nEND

Expected Output: A parameter control with Sales and Profit options appears. Selecting either option changes the measure shown in the chart.

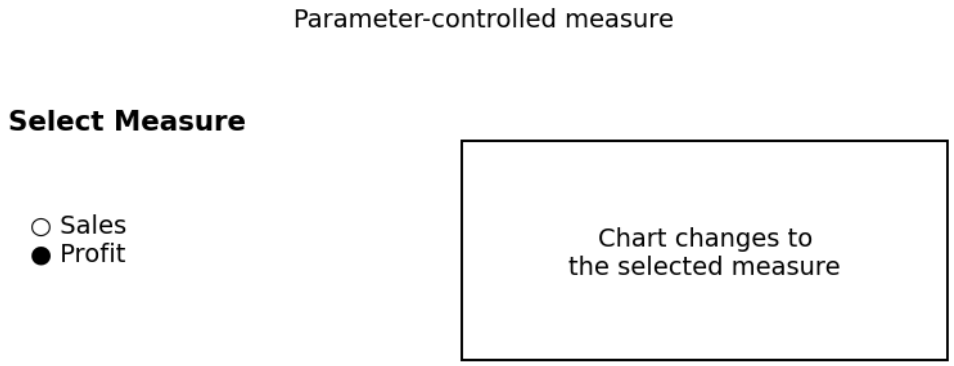


Diagram: illustrative expected output

Q7. c) Use URL actions for interactivity.

Steps

127. Open the dashboard.
128. Click Dashboard → Actions.
129. Select Add Action → Go to URL.
130. Enter the required URL.
131. Choose trigger, such as Select → OK.

Expected Output: Selecting a mark triggers the configured URL action and opens the selected web address.

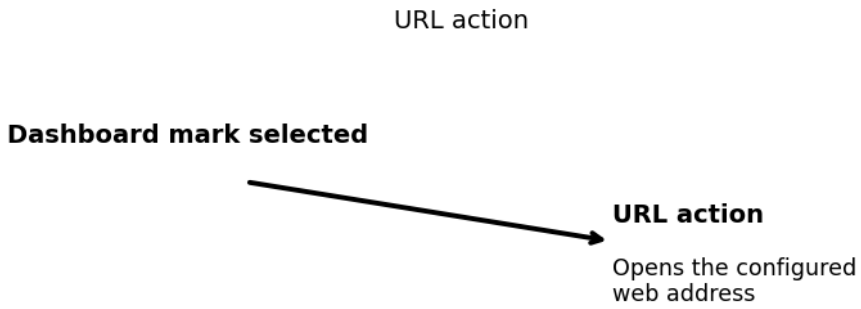


Diagram: illustrative expected output

Q8. a) Apply quick table calculations (running total, percent of total).

Steps

132. Running Total
133. Create a chart using Order Date and Sales.
134. Right-click SUM(Sales).
135. Select Quick Table Calculation → Running Total.
136. Percent of Total
137. Right-click SUM(Sales).
138. Select Quick Table Calculation → Percent of Total.

Expected Output: The running-total view accumulates Sales over the date sequence. The percent-of-total view shows each mark's contribution to the total.

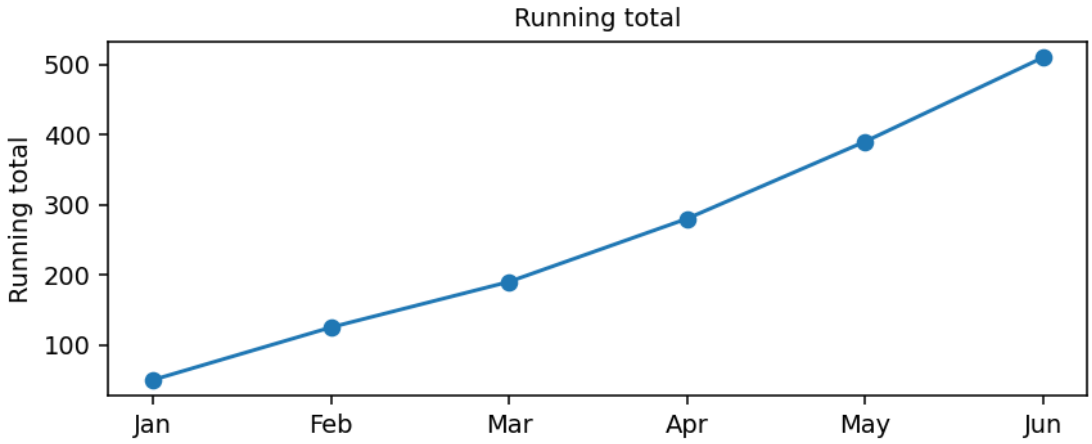
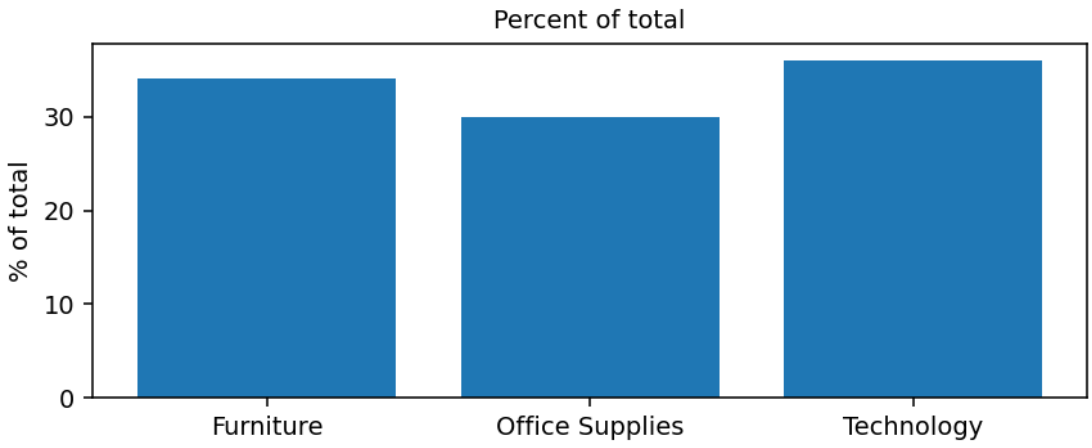


Diagram: illustrative expected output



Expected output for percent of total

Q8. b) Create moving averages and cumulative metrics.

Steps

139. Moving Average
140. Right-click Sales.
141. Select Quick Table Calculation → Moving Average.
142. Cumulative Metric
143. Right-click Sales.
144. Select Quick Table Calculation → Running Total.

Expected Output: A moving-average line smooths fluctuations in Sales. A cumulative metric increases or stays level as values are accumulated across the sequence.

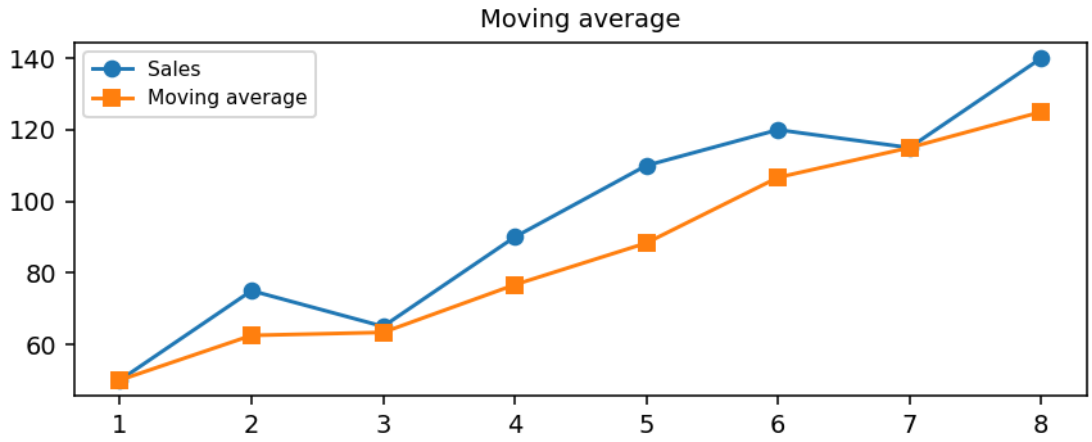


Diagram: illustrative expected output

Q9. a) Create basic calculated fields using formulas.

Steps

145. Click Analysis → Create Calculated Field.
146. Name it Profit Ratio.
147. Enter the formula shown below.
148. Click OK.

Formula / text: [Profit] / [Sales]

Expected Output: A new calculated field named Profit Ratio appears in the Data pane and can be used in a view.

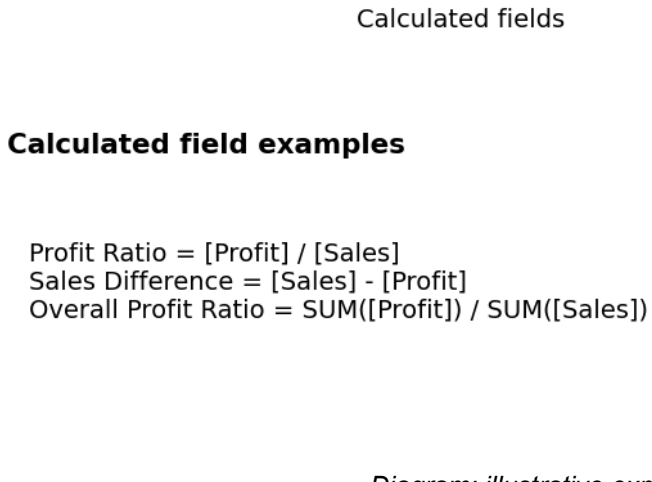


Diagram: illustrative expected output

Q9. b) Perform row-level calculations.

Steps

149. Create a calculated field.
150. Name it Sales Difference.
151. Enter the formula shown below.
152. Click OK.

Formula / text: [Sales] - [Profit]

Expected Output: The Sales Difference field calculates Sales minus Profit for each underlying record.

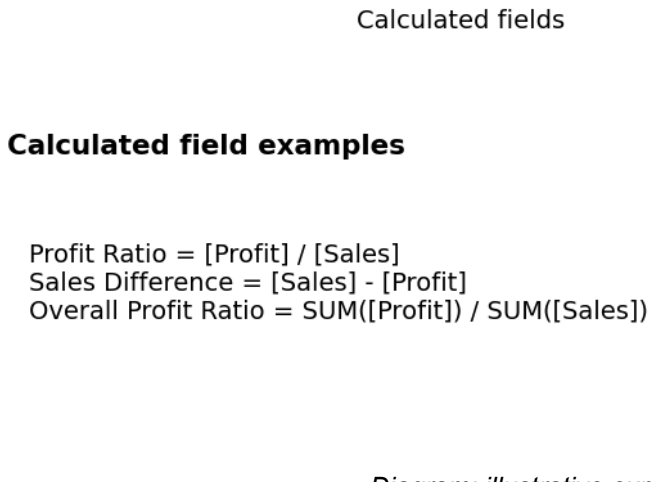


Diagram: illustrative expected output

Q9. c) Perform aggregate-level calculations.

Steps

153. Create a calculated field.
154. Name it Overall Profit Ratio.
155. Enter the formula shown below.
156. Click OK.

Formula / text: SUM([Profit]) / SUM([Sales])

Expected Output: The Overall Profit Ratio field calculates total Profit divided by total Sales for the marks in the view.

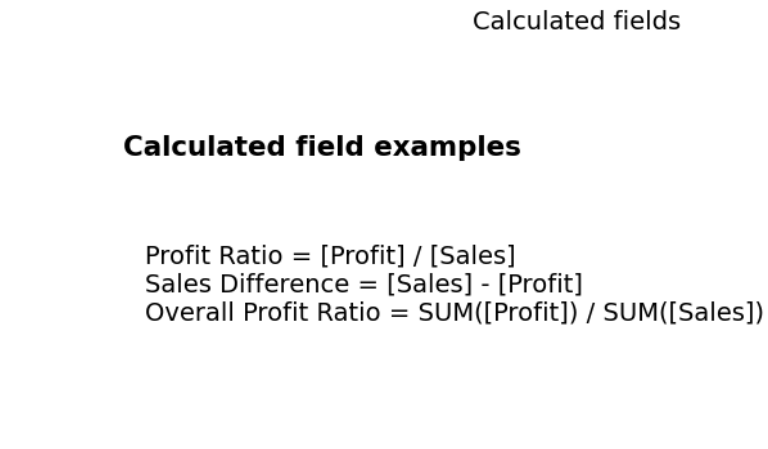


Diagram: illustrative expected output

Note: The sample formulas are copied from your uploaded practical document. If Sales is zero in a row, Profit Ratio may return a null or error depending on the data and Tableau handling.

Q10. a) Design dashboards for multiple devices.

Steps

157. Open the dashboard.
158. Find Device Preview in the Dashboard pane.
159. Preview Desktop, Tablet and Phone layouts.
160. Adjust the layout for each device.

Expected Output: The dashboard preview shows desktop, tablet and phone layouts, with content resized or rearranged for each screen.

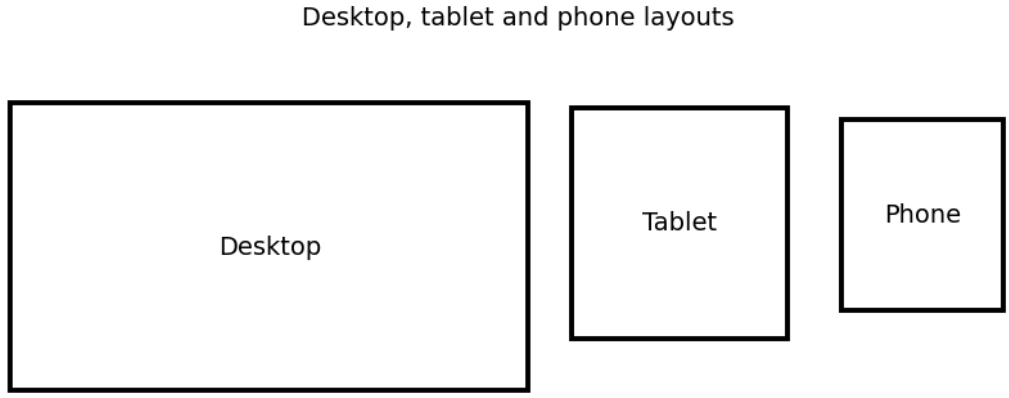


Diagram: illustrative expected output

Note: Device Preview availability and exact controls can vary by Tableau version.

Q10. b) Build a complete data story using the Story feature.

Steps

161. Click New Story.
162. Set the story size.
163. Add Sales by Category.
164. Click New Story Point.
165. Add Sales by Region.
166. Add another story point for Profit Analysis.
167. Give each story point a suitable title.

Expected Output: A story contains sequential story points for Sales by Category, Sales by Region and Profit Analysis, each with a suitable title.

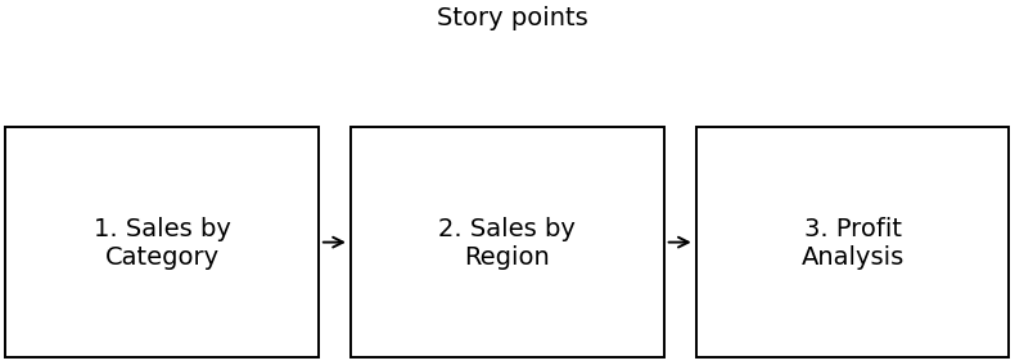


Diagram: illustrative expected output

Q11. a) Apply data blending between two sources.

Steps

168. Connect to the first data source.
169. Click Data → New Data Source.
170. Connect to the second data source.
171. Open a worksheet.
172. Use fields from both sources.
173. Identify the common linking field.

Expected Output: The view uses fields from both data sources, linked by the common field. Data blending combines data from different data sources.

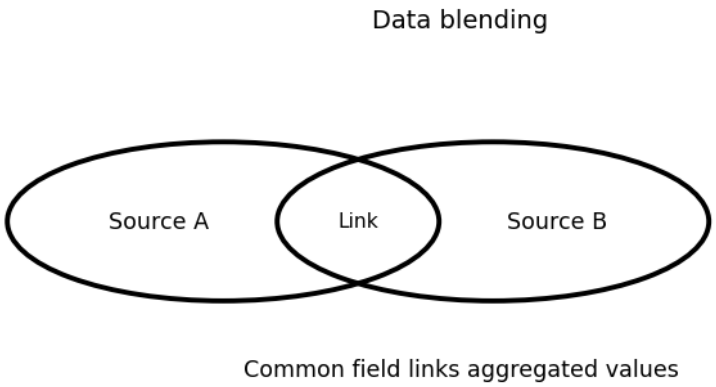


Diagram: illustrative expected output

Q11. b) Implement different types of filters (dimension, measure, date).

Steps

174. Dimension: Category → Filters.
175. Measure: Sales → Filters.
176. Date: Order Date → Filters.
177. Select the required values → OK.

Expected Output: The view is restricted according to the selected category, sales range and date period.

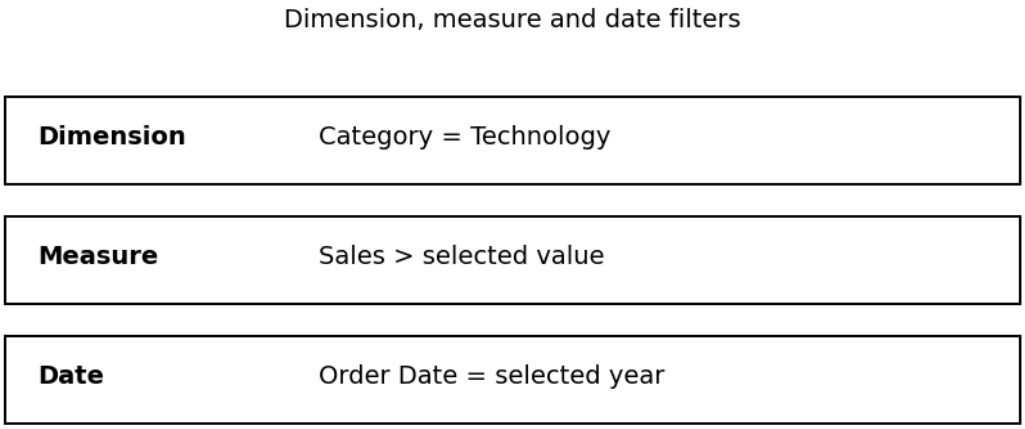


Diagram: illustrative expected output

Q11. c) Create dynamic titles and labels using table calculations.

Steps

178. Create a visualization with a table calculation.
179. Add the calculation to the required measure.
180. Go to Marks → Label → Show Mark Labels.
181. Edit the title → Insert the relevant field.
182. Click OK.

Expected Output: The title and mark labels update to reflect the calculated data in the current view.

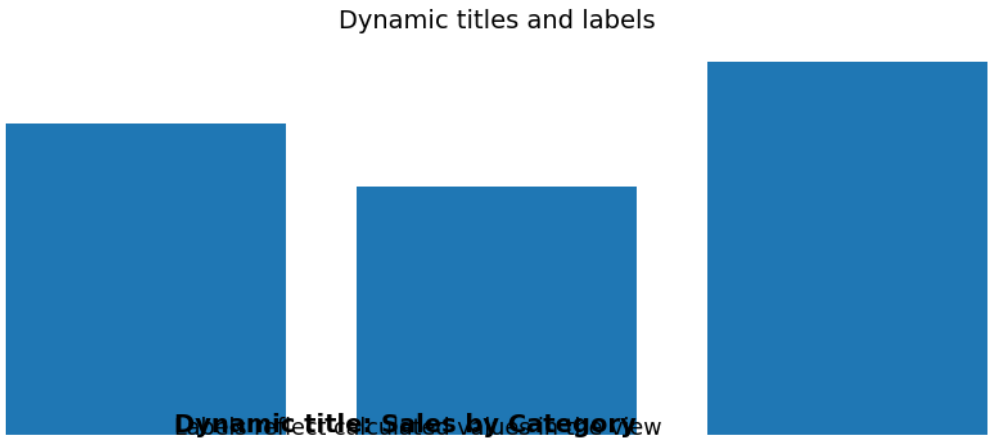


Diagram: illustrative expected output

Q12. a) Add and interpret trend lines (linear, exponential).

Steps

183. Create a suitable chart using numerical fields.
184. Click Analysis → Trend Lines → Show Trend Lines.
185. Select the required trend model.
186. Common models: Linear, Exponential, Logarithmic, Polynomial, Power.

Expected Output: The chart displays the data and a trend line. An upward line indicates a general increase; a downward line indicates a general decrease.

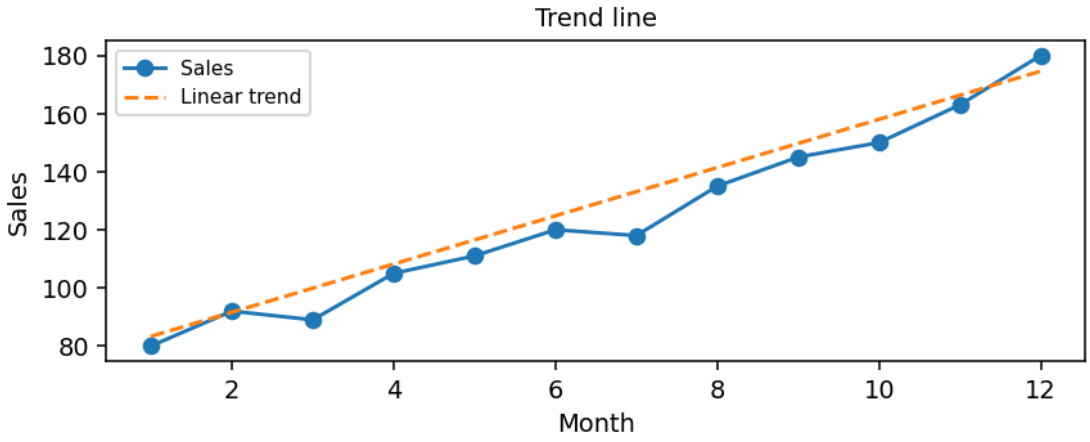


Diagram: illustrative expected output

Q12. b) Customize and analyze trend models.

Steps

187. Right-click the trend line.
188. Select Describe Trend Line.
189. Examine the equation, R-squared and statistical information.

Expected Output: Tableau displays trend-model details, including the equation and R-squared where supported. R^2 closer to 1 generally indicates a stronger fit to the observed data.

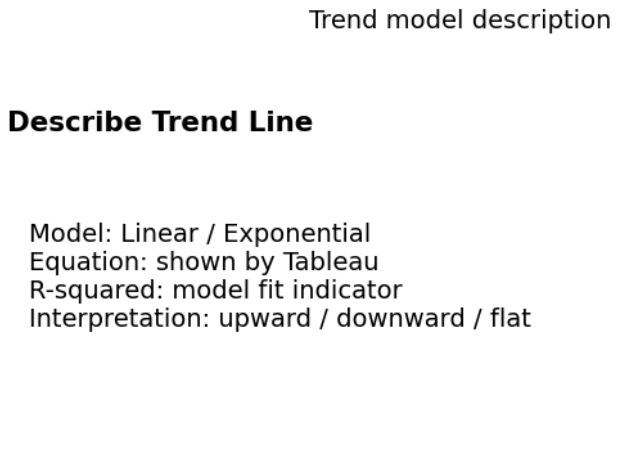


Diagram: illustrative expected output

Quick Matching Check

This document follows your uploaded tableau-1.docx structure: Q1(a–c), Q2(a–b), Q3(a–c), Q4(a–b), Q5(a–c), Q6(a–b), Q7(a–c), Q8(a–b), Q9(a–c), Q10(a–b), Q11(a–c), and Q12(a–b). Each sub-question is separated and includes an expected output plus a diagram.

Reminder: Diagrams are sample illustrations to help you recognize the expected result in the practical exam. The exact chart values and interface appearance can differ on your computer.